Name: _____ Date: _____ Period: _____

Unit 1- Practice Quiz # 1

1) Find the value of n that makes this statement true:

$$\mathbf{3^3 \cdot 3^n = 3^7}$$

2) Find the value of n that makes this statement true:

$$\frac{\mathbf{5^{12}}}{\mathbf{5^n}} = \mathbf{5^4}$$

3) What is the area of the rectangle below?



$9x^2$

$12x^3$

4) Students were asked to simplify the expression using laws of exponents:

$$\mathbf{7^5 \cdot 7}$$

Who was correct?

a) Jay wrote 49^5 c) Kayla wrote 49^6

b) Dillon wrote 7^6 d) Alex wrote 7^5

5) Students were asked to simplify the expression using laws of exponents:

$$\frac{\mathbf{9^{10}}}{\mathbf{9^2}}$$

Who was correct?

a) Jay wrote 1^5 c) Kayla wrote 9^5

b) Dillon wrote 1^8 d) Alex wrote 9^8

6) Sal multiplied the following: $(12a^5)(3a) = 36a^4$

Is Sal correct? Explain how you determined your answer.

7) Elicia says that $(-4)(-4)(-4)(-4)(-4)(-4)(-4)(-4) = -4^8$. Is Elicia correct in her notation? Why or why not?

8) Simplify the following. Leave you answer in exponential notation.

$$\mathbf{3^5 \cdot \frac{3^6}{3^2}}$$

9) Given the expression below, what is an equivalent expression in exponential form?

$$\mathbf{-5 \cdot -5 \cdot n \cdot n \cdot n \cdot n}$$

a) $(-5)^2n^4$

b) -5^3n^4

c) $(-5n^2)^4$

d) -5^2n^4

10) Simplify the expression using the laws of exponents. Leave your answer in exponential form

$$\mathbf{\frac{(3^4 \cdot 3^6)}{3^2}}$$